



Reflect with us - Week 6



This assignment will not be graded and is only for practice.

Suppose the matrix representation of a linear transformation $T: R^3 \rightarrow R^3$ with respect to ordered bases

$$\beta = \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\} \quad \text{for the domain and}$$

$$\gamma = \{(1, 0, 0), (0, 1, 0), (1, 0, 1)\} \quad \text{for the range,}$$

is $I_{3 \times 3}$, i.e., the identity matrix of order 3.

Let AA denote the matrix representation of the linear transformation T with respect to the standard ordered basis of R^3 for both domain and range. Which of the following are true?

a. $A = I_{3 \times 3}$, i.e., identity matrix of order 3.

b. AA is a singular matrix.

c. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

d. $\det(A) = 1$

e. $\det(A) = -1$

f. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

Solution:

Observe that, in question, you are asked to find the matrix representation of T with respect to the standard ordered bases of R^3 for both domain and range. So you have to find the image of each element of the standard ordered basis of R^2 via the linear transformation T .

Note: $I_{3 \times 3}$ is matrix representation of the T with respect to the ordered bases β and γ .

Step 0:

What is $T(1, 0, 1)$? (Enter your answer in the form of a row vector in R^3)

Note: The answer will be of the form (a, b, c) where a, b and c are in R

No, the answer is incorrect.

Score: 0

Feedback:

From the given data, we can calculate

$$T(1, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 0(1, 0, 1) = (1, 0, 0)$$

Accepted Answers:



(Type: String) (1, 0, 0)

(Type: String) (1,0,0)

1 point

What is $T(0, 1, 0)T(0, 1, 0)$?**No, the answer is incorrect.****Score: 0****Feedback:**

From the given data, we can calculate

$$T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(1, 0, 1) \quad T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(1, 0, 1)$$

Accepted Answers:

(Type: String) (0, 1, 0)

(Type: String) (0,1,0)

1 point

What is $T(0, 0, 1)T(0, 0, 1)$?**No, the answer is incorrect.****Score: 0****Feedback:**

From the given data, we can calculate

$$T(0, 0, 1) = 0(1, 0, 0) + 0(0, 1, 0) + 1(1, 0, 1) \quad T(0, 0, 1) = 0(1, 0, 0) + 0(0, 1, 0) + 1(1, 0, 1)$$

Accepted Answers:

(Type: String) (1, 0, 1)

(Type: String) (1,0,1)

1 point

Step 1: Step 1:

Now we have to find the image of $(1, 0, 0)(1, 0, 0)$ via the linear transformation T . We know the image of each element of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\} \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$. So at first we have to write $(1, 0, 0)(1, 0, 0)$ in terms of the elements of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\} \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$.

$$(1, 0, 0) = \alpha_1(1, 0, 1) + \beta_1(0, 1, 0) + \gamma_1(0, 0, 1) \quad (1, 0, 0) = \alpha_1(1, 0, 1) + \beta_1(0, 1, 0) + \gamma_1(0, 0, 1)$$

(0, 0, 1)

What is the value of α_1 ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 1

1 point

What is the value of β_1 ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

What is the value of γ_1 ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

1 point

Now, can you find $T(1, 0, 0)T(1, 0, 0)$?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

From the given data, we can calculate

$$T(1, 0, 0) = T(1(1, 0, 1) + 0(0, 1, 0) - 1(0, 0, 1))T(1, 0, 0) = 1T(1, 0, 1) + 0T(0, 1, 0) - 1T(0, 0, 1)$$

$$T(1, 0, 0) = T(1(1, 0, 1) + 0(0, 1, 0) - 1(0, 0, 1))T(1, 0, 0) = 1T(1, 0, 1) + 0T(0, 1, 0) - 1T(0, 0, 1)$$

Here we know the images of $(1, 0, 1)$, $(0, 1, 0)$, $(1, 0, 1)$, $(0, 1, 0)$ and $(0, 0, 1)$.

$$T(1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) - 1(1, 0, 1) = (0, 0, -1)T(1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) - 1(1, 0, 1) = (0, 0, -1)$$

Accepted Answers:

Yes

No

Step 2:Step 2:

Now we have to find the image of $(0, 1, 0)$ via the linear transformation T . We know the image of each element of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$. So at first we have to write $(0, 1, 0)$ in terms of the elements of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$.

$$(0, 1, 0) = \alpha_2(1, 0, 1) + \beta_2(0, 1, 0) + \gamma_2(0, 0, 1)(0, 1, 0) = \alpha_2(1, 0, 1) + \beta_2(0, 1, 0) + \gamma_2(0, 0, 1)$$

What is the value of α_2 ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

What is the value of β_2 ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

What is the value of $\gamma_2 \gamma_2$?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

1 point

Now, can you find $T(0, 1, 0)T(0, 1, 0)$?☐ Yes☐ No**No, the answer is incorrect.****Score: 0****Feedback:**

$$T(0, 1, 0) = T(0(1, 0, 1) + 1(0, 1, 0) + 0(0, 0, 1))T(0, 1, 0) = T(0(1, 0, 1) + 1(0, 1, 0) + 0(0, 0, 1))$$

$$T(0, 1, 0) = 0T(1, 0, 1) + 1T(0, 1, 0) + 0T(0, 0, 1)T(0, 1, 0) = 0T(1, 0, 1) + 1T(0, 1, 0) + 0T(0, 0, 1)$$

$$T(0, 1, 0) = (0, 1, 0)T(0, 1, 0) = (0, 1, 0)$$

Accepted Answers:

Yes

No

Step 3:Step 3:

Now we have to find the image of $(0, 0, 1)(0, 0, 1)$ via the linear transformation T . We know the image of each element of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$. So at first we have to write $(0, 0, 1)(0, 0, 1)$ in terms of the elements of the basis $\{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$.

$$(0, 0, 1) = \alpha_3(1, 0, 1) + \beta_3(0, 1, 0) + \gamma_3(0, 0, 1)(0, 0, 1) = \alpha_3(1, 0, 1) + \beta_3(0, 1, 0) + \gamma_3(0, 0, 1)$$

What is the value of α_3 ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

What is the value of β_3 ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

What is the value of γ_3 ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

1 point

Now, can you find $T(0, 0, 1)T(0, 0, 1)$?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Feedback:

$$T(0, 0, 1) = T(0(1, 0, 1) + 0(0, 1, 0) + 1(0, 0, 1))T(0, 0, 1) = T(0(1, 0, 1) + 0(0, 1, 0) + 1(0, 0, 1))$$

$$T(0, 0, 1) = 0T(1, 0, 1) + 0T(0, 1, 0) + 1T(0, 0, 1)T(0, 0, 1) = 0T(1, 0, 1) + 0T(0, 1, 0) + 1T(0, 0, 1)$$

$$T(0, 0, 1) = (1, 0, 1)T(0, 0, 1) = (1, 0, 1)$$

Accepted Answers:

Yes

No

Step 4: Step 4: Now we have to find out the matrix AA . For that we have to write down the images with respect to the standard ordered basis.

$$T(1, 0, 0) = 0(1, 0, 0) + 0(0, 1, 0) - 1(0, 0, 1)T(1, 0, 0) = 0(1, 0, 0) + 0(0, 1, 0) - 1(0, 0, 1)$$

$$T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(0, 0, 1)T(0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(0, 0, 1)$$

$$T(0, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 1(0, 0, 1)T(0, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 1(0, 0, 1)$$

Write the coefficients column wise to get the matrix AA .

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Step 5: Step 5:

In this step, you have to find the determinant of the matrix AA .

1 point

Can you find the $\det(A)$?

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes

No

- Expanding with respect to the first row:

$$\det(A) = 0(1 - 0) - 0(0 - 0) + 1(0 + 1)\det(A) = 0(1 - 0) - 0(0 - 0) + 1(0 + 1)$$

- Expanding with respect to the first column:

$$\det(A) = 0(1 - 0) - 0(0 - 0) - 1(0 - 1)\det(A) = 0(1 - 0) - 0(0 - 0) - 1(0 - 1)$$

- You can expand with respect to any other rows or column.

Enter the value of *determinant of the matrix A* determinant of the matrix A

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Your score is: 0/17